Heat Capacity of an Interacting Spin System

Source: D. Tong, Statistical Physics, Problem Sheet 1, Question 4.

Consider a system of N interacting spins. At low temperatures, the interactions ensure that all spins are either aligned or anti-aligned with the z axis, even in the absence of an external field. At high temperatures, the interactions become less important and spins can point in either $\pm\hat{z}$ direction. If the heat capacity takes the form

$$C_V = C_{\max}\left(\frac{2T}{T_0} - 1\right) \quad \text{for } \frac{T_0}{2} < T < T_0 \quad \text{and} \quad C_V = 0 \text{ otherwise,} \tag{1.1}$$

determine $C_{\max}$.

Solution. As we have seen in the previous question, the maximum entropy of such a system is given by $Nk_B \ln 2$, and the minimum entropy is 0. By the definition of entropy, we find

$$\Delta S = Nk_B \ln 2 = \int_0^\infty \frac{C_V dT}{T}. \tag{1.2}$$

Take the heat capacity to be of the form given in the question and evaluate the integral, we find

$$\Delta S = C_{\max}(1 - \ln 2) \implies C_{\max} = \frac{Nk_B \ln 2}{1 - \ln 2}. \tag{1.3}$$